



Draft-genome analysis provides insights into the virulence properties and genome plasticity of *Vibrio fluvialis* organisms isolated from shrimp farms and Turag river in Bangladesh

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Abstract

Vibrio fluvialis is an opportunistic waterborne and seafood-borne enteric pathogen capable of causing severe diarrhea leading to death. This pathogen is endemic to Bangladesh, a country which is a major producer of cultured shrimp and wild-caught prawns. In this study, we carried out whole-genome sequencing of three *V. fluvialis* organisms isolated from shrimp farm and river sediment showing strong pathogenic characteristics in vivo and in vitro and compared their genomes against other *V. fluvialis* and related pathogenic species to glean insights into their potential as pathogens. Numerous virulence-associated genes including hemolysins, cytotoxins, three separate Type IV pili, Types II and VI secretion systems, biofilm, and the *V. cholerae* pathogenesis regulating gene, *toxR*, were identified. Moreover, we found strain S-10 to have the propensity to acquire antibiotic resistance genes through horizontal gene transfer. These findings indicate that shrimp farms and rivers could be potential sources of *V. fluvialis* organisms which are an infection threat of public health concern.

Keywords *Vibrio fluvialis* · Genomics · Virulence factors · Mobile genetic elements · Pathogenicity · Bangladesh

Introduction

Vibrio fluvialis is an opportunistic waterborne and seafood-borne pathogen that is typically associated with enteric infections causing acute diarrhea which can lead to death and is also capable of causing extraintestinal infections such as cellulitis, necrotizing fasciitis, tissue necrosis, primary sepsis, and suppurative cholangitis (Igbinosa and Okoh 2010). Since this organism was first isolated as a “Vibrio-like

organism EF-6” from more than 500 patients with diarrhea in Bangladesh between October 1976 and June 1977 (Huq et al. 1980), several outbreaks of *V. fluvialis* mediated diarrhea have occurred infrequently in flood-prone developing countries of South Asia that generally lack good sanitation and hygiene practices, particularly in Bangladesh. A 2010 study conducted on the residents of Pakhirala village on an island in the Sunderbans (Bangladesh) found that 91.2% of residents were infected with *V. fluvialis* in the aftermath of cyclone Aila in 2009 (Bhattacharjee et al. 2010). *V. fluvialis* was also associated with outbreaks of diarrhea during the 1998 flood in Bangladesh (Tanabe et al. 1999). Given the history of outbreaks of diarrhoeal infections caused by *V. fluvialis*, this pathogen is of great public health concern in this country.

Bangladesh is a major producer of both farmed marine shrimp and wild-caught and cultured freshwater prawns. Given that *V. fluvialis* has been documented to be transmitted through seafood, these shrimp farms and rivers could represent a source of disease-causing organisms. Moreover, the overuse or misuse of antibiotics in aquaculture may lead to toxic effects and the proliferation of antimicrobial resistance in pathogenic bacteria (Mohan et al. 2022). Thus, in

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one of our previous studies, we isolated *Vibrio* species from river and shrimp farm sediments and found that *V. fluvialis* organisms showed seasonal trends in abundance. Sixty-seven percent of strains were isolated in the month of September, 33% in February, and none in January, March, July, and December. There was also variation in the *V. fluvialis* prevalence in river sediment vs shrimp farm with 19.4% of Turag river samples containing *V. fluvialis* and only 5.13% of shrimp farm samples containing *V. fluvialis*. One-third of river samples did not contain any *Vibrio* species while most of the rest contained *V. parahaemolyticus*. Among other species *V. alginolyticus*, *V. cholerae*, *V. mimicus* being the most prevalent. Only two samples from shrimp farms did not contain *Vibrio* species, with more than 90% containing *V. parahaemolyticus* (lacking virulence genes) and *V. alginolyticus*.

In another recent study of ours (Biswas et al. 2020), we investigated the phenotypic virulence properties of these *V. fluvialis* organisms. Culture filtrates and crude proteins prepared from shrimp farm and river sediment *V. fluvialis* isolates showed fluid accumulation in rabbit ileal loop assay, hemolysis of sheep red blood cells, rounding of BHK-21, HeLa, and MDCK cells in cell culture assay, hind limb paralysis and death of mice in mouse lethality assay and morphological changes in mouse neuronal cell assay (Biswas et al. 2020). To elaborate on our research findings, in this study, we carried out whole-genome sequencing of three of the *V. fluvialis* strains (S-5, S-9, and S-10), with an aim to identify the genetic elements associated with pathogenesis, antibiotic resistance, and genome plasticity. Comparative analysis was also carried out with 53 *V. fluvialis* genomes available at NCBI to determine whether these strains have the genetic elements that are responsible for virulence in pathogenic *V. fluvialis* and how our strains are related to different clinical and environmental isolates.

Materials and methods

Strain selection

From our previous study (Biswas et al. 2020), three strains showing the most pathogenic potential were chosen for genomic analysis: S-5 obtained from the downstream sediment of Turag river located near Dhaka, S-9, and S-10 obtained from the sediment of shrimp farms in Satkhira located in the southwestern part of Bangladesh.

Genome sequencing, assembly, and comparative analysis

Library preparation and sequencing of three strains, S-5, S-9, and S-10 were conducted at NSU Genome Research Institute (NGRI), North South University, Dhaka, Bangladesh.

Genomic DNA was extracted (GeneJetDNA Purification Kit, Thermo Fisher) and libraries were prepared using the Nextera DNA Library Preparation Kit (Cat. No. FC-121-1030) following the manufacturer's guidelines and sequencing was carried out using the Illumina MiSeq platform. Good quality paired-end reads ($Q \geq 30$) were assembled de novo using SPAdes (v3.12) (Bankevich et al. 2012). After removing contigs shorter than 200 bp, RAST (Aziz et al. 2008) was used for subsystem annotation and analysis. PATRIC (Wattam et al. 2014) was used to annotate virulence factors, while antimicrobial resistance genes, prophage sequences, plasmid sequences, and insertion sequences were annotated using CARD (Alcock et al. 2020) and ResFinder (Bortolaia et al. 2020), PHASTER (Arndt et al. 2016), PLSDB (Galata et al. 2019), and ISFinder (Siguier et al. 2006), respectively.

Pan and core genome analysis was carried out using ROARY (Page et al. 2015). The core genome alignment was used to generate an unrooted maximum-likelihood phylogenetic tree via RaxML (Stamatakis 2014), with a bootstrap value of 200. Hierarchical clustering based on the presence and absence of accessory genes was performed using the PVclust R package (<https://cran.r-project.org/web/packages/pvclust/index.html>).

Results and discussion

Genomic features and phylogenetic analysis of *V. fluvialis* genomes

The estimated draft-genome sizes of the three strains ranged between ~4.61 and 4.82 M bp with GC % of ~50.0% (Online Resource 1). Analysis of the pan-genome, the entire gene set of all strains, revealed that the number of novel genes reduces, but does not plateau with the gradual addition of more genomes, indicating that the 50–56 sequenced strains have captured the majority, but not all of the variations within *V. fluvialis* genomes (Fig. 1). The total number of genes in the pan-genome is 13,837, with 1495 core genes.

A total of 5072 unique genes were found, representing 36.66% of the total gene count, indicating a high level of diversity in the pan-genome of *V. fluvialis*. However, 2211, or 43.60%, of these unique genes encode hypothetical proteins. All three of our strains were found to possess unique genes, with S-5 having 4, S-9 having 6, and S-10 having the most at 96 genes (Online Resource 1). Strain S-10 contains interesting unique genes encoding 16 mobile genetic elements including phage elements and transposases as well as a Csy3 family CRISPR-associated protein. Moreover, it contains putative virulence-related genes encoding O-antigen export system permease protein RfbD, UDP-4-amino-4-deoxy-L-arabinose-oxoglutarate aminotransferase *arnB*, dTDP-4-dehydrorhamnose 3,5-epimerase

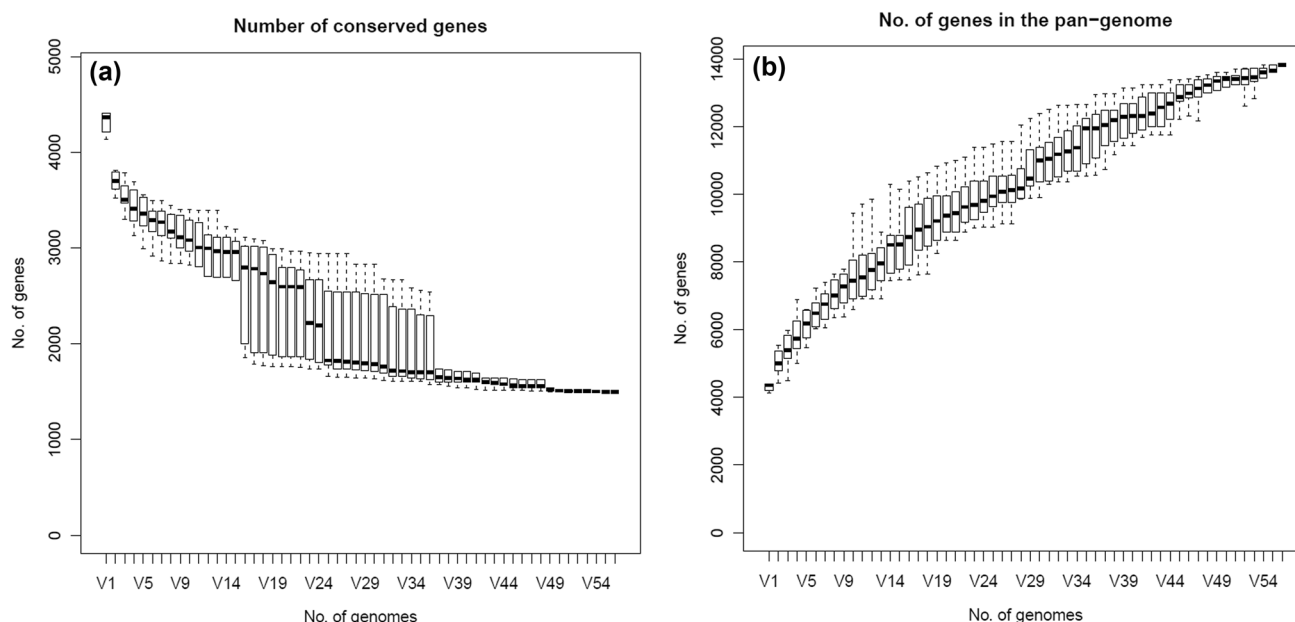


Fig. 1 Relationship between gene count and genome size: **a** the size of the core genome (the number of genes that are common among all samples) and **b** the size of the pan-genome (the sum of all different genes identified in all samples) were determined when between 2 and

56 genomes were randomly combined with 1000 permutations. The boxplots represent the median and interquartile ranges of 1000 permutations, while the whisker plots show the extreme values

RfbC, Lipid carrier: UDP-*N*-acetylgalactosaminyl transferase/Alpha-1,3-*N*-acetylgalactosamine transferase PglA, and dTDP-rhamnosyl transferase RfbF. All these genes are related to the synthesis of bacterial O-antigen and are virulence factors in other pathogenic species such as *Escherichia coli*, and *Vibrio cholerae*.

Phylogenetic analysis (Fig. 2) based on SNP distribution in the core genome sorted 56 *V. fluvialis* strains into three distinct clades, with clade C possessing most of the strains (~90%) and a more complex subclade system in comparison to clades A and B. Except for NBRC_103150 and FDAARGOS_104, the earliest (1981) Bangladeshi *V. fluvialis* genomes available, all other strains from Bangladesh were grouped into clade C. Within clade C, strain S-10 clustered away from S-5 and S-9. These two clustered closer to a subclade containing the majority of the clinical strains with the prefix “GCCIn”- denoting Bangladeshi *V. fluvialis* strains recently sequenced by icddr,b.

Cluster dendrogram (Online Resource 8) based on the presence/absence of the accessory gene pool revealed a more complex pattern of clustering, with no clearly definable clades. However, similar to the core genome phylogenetic tree, strain S-10 clustered separately from S-5 and S-9. Moreover, both groups have retained their nearest neighbors from the core genome tree in the accessory gene-based clustering, with S-10 clustering close to GCCIn31 and GCCIn56 and S-5 and S-9 clustering close to Ak_1296.A2.1 and FDAARGOS_100. This clustering pattern was unexpected

given that strains S-9 and S-10 were isolated from similar locations (shrimp farm sediment in Satkhira), whereas S-5 was isolated from the sediment of Turag river. Sixty-six genes were found to be unique to S-5 and S-9, with a majority of these encoding hypothetical proteins, although there are multiple genes that encode mobile genetic elements, mostly prophages. Moreover, they both possess the IncP-type DNA relaxase TraI, an enzyme that catalyzes site-specific cleavage and joining of ssDNA that is typically found in Inc-type plasmids (Pansegrau et al. 1993). They also possess a DNA-cytosine methyltransferase, an enzyme that catalyzes the methylation of cytosine nucleotides, playing a key role in the epigenetic control of gene expression. Furthermore, they possess a gene encoding capsular polysaccharide biosynthesis protein, which is involved in the synthesis of the eponymous capsular polysaccharide which is a major virulence factor in *Streptococcus pneumoniae* (Saksouk et al. 2005). However, the two strains that were isolated from shrimp farm sediments, S-9 and S-10, did not have any genes that were unique to them. Taken altogether, the presence of acquired virulence genes that are unique to these strains and the numerous mobile genetic elements show the potential of *V. fluvialis* to acquire virulence genes from other pathogenic species through horizontal gene transfer.

Tree scale: 0.01

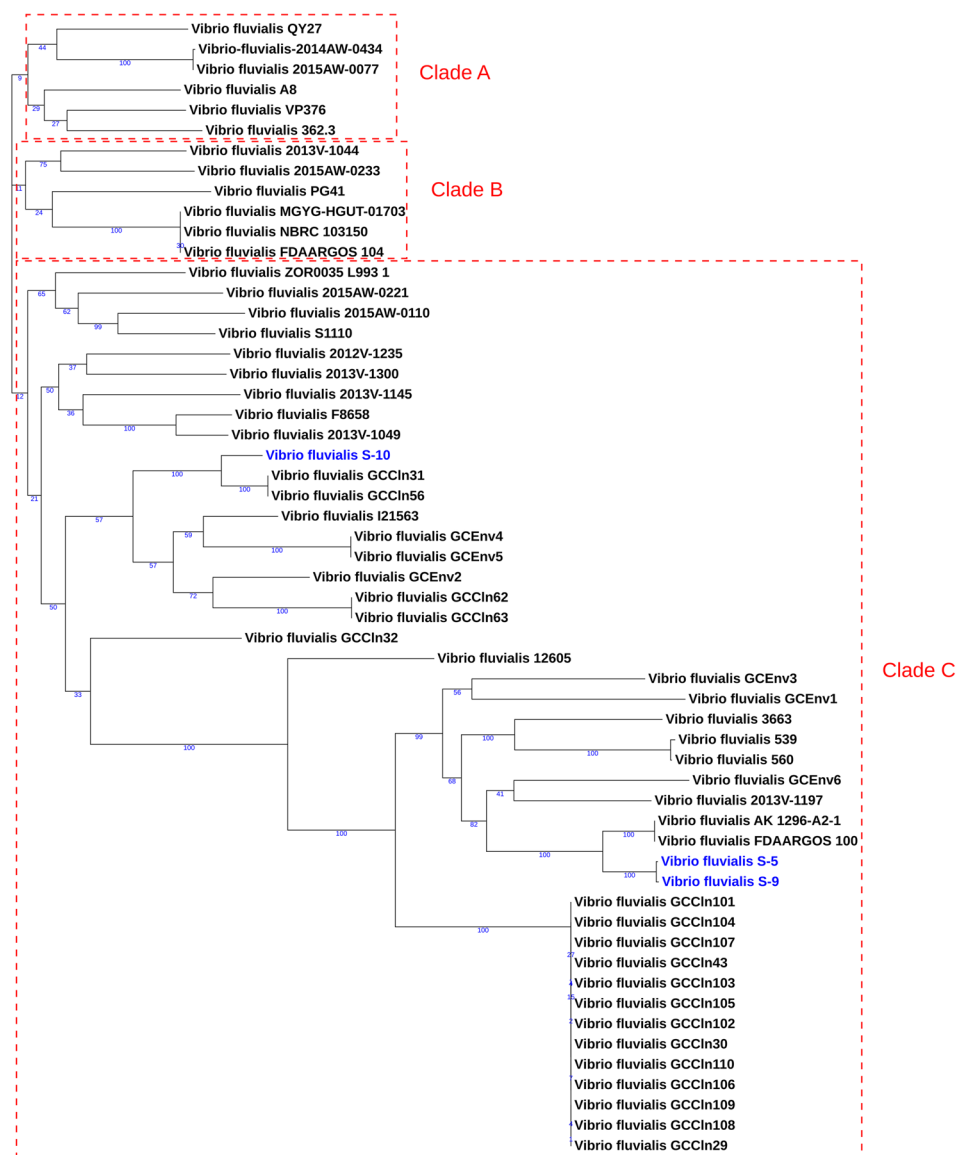


Fig. 2 The core genome phylogenetic tree, an unrooted Maximum Likelihood tree generated using RAxML with the GAMMA + P-Invar Model. Two hundred bootstraps were performed to ascertain the probability of each clustering pattern and the best tree which displays

the maximum-likelihood clustering pattern is shown here. The three *V. fluvialis* strains included in this study are in blue (colour figure online)

Analysis of virulence factors

In our previous study, molecular analysis of isolates showed the absence of classical virulence factors such as *ctx*, *tdh*, *trh*, *stx1*, and *stx2* (Biswas et al. 2020). However, culture filtrates and crude proteins prepared from these organisms showed numerous virulence properties in vitro implying the presence of different virulence genes

responsible for enterotoxin, hemolysin, cytotoxin, and neurotoxin production. Through the use of genome sequencing, we investigated the presence of different virulence genes in *V. fluvialis* isolates in Bangladesh and their potential role in pathogenesis.

Factors related to hemolysis, cytotoxicity, and enterotoxigenicity

Several genes encoding hemolysins and associated proteins were found in all three strains (Online Resource 2), including thermostable hemolysin delta, thermolabile hemolysin precursor, *V. fluvialis* hemolysin vfh (encoded by the *hlyA* gene), and hemolysin secretion protein HylB. The *V. fluvialis* hemolysin has haemagglutinating, permeability-enhancing, and hemorrhagic activities that enhance hemolysis and enterotoxigenicity (Kothary et al. 2003). It may also enable cell vacuolation as it does in *V. cholerae* (Figueroa-arredondo et al. 2001). Meanwhile, the thermostable hemolysin gene delta is associated with, but not solely responsible for, enterotoxigenicity, hemolysis, and cytotoxicity in the related species *V. parahaemolyticus* (Raghunath 2014). All three strains also possess the *hupO* gene which encodes a hemin-binding outer membrane protein and hemoglobin receptor (Online Resource 2). This protein enhances virulence by stimulating hemolysin production (Ahn et al. 2005). Moreover, we found our strains to contain components of the Cytolysin and Lipase operon in *Vibrio* (Online Resource 2) which is associated with the cytotoxic effects of *V. cholerae* (Ogierman et al. 1997). In *V. cholerae*, the *hlyA* encoded *V. cholerae* cytolysin (VCC) is associated with cytotoxic effects, while the product of the *hlyB* gene is involved in the secretion of VCC (Kathuria and Chattopadhyay 2018). Another gene encodes the protease PrtV. *V. cholerae* PrtV has been shown to induce cytotoxic effects on human intestinal cell line HCT8 and degrade extracellular matrix components fibronectin and fibrinogen, and plasminogen. (reference: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2613228/>)

The varied hemolysins, toxins, and related genes discussed above are likely responsible for the hemolysis, enterotoxigenicity, and cytotoxicity demonstrated in our previous study (Biswas et al. 2020) and make these strains likely to be able to cause similar effects in the human body. This shows the potential threat posed by *V. fluvialis* organisms present in shrimp farms and rivers. Interestingly, while absent in our strains, 16 *V. fluvialis* strains including the Bangladeshi FDAARGOS_104, NBRC 103150, and GCEnv2 possess *ace* (accessory cholera enterotoxin), *zot* (zona occludens toxin), *toxR* (Transcriptional activator), and *toxS* (Transmembrane regulatory protein) genes (Online Resource 2) belonging to the cholera toxin subsystem found on the CTXφ phage in *V. cholerae*. The presence of genes encoding the *zot* and *ace* toxins is a novel discovery in *V. fluvialis*. Moreover, the presence of this phage indicates the potential for the transfer of genetic elements encoding virulence factors between *Vibrio* species.

Type IV Pili, flagella, and other factors associated with attachment and colonization of shellfish and the human intestine

The ability of *Vibrio* spp. to attach themselves to and colonize shellfish and the human intestine is crucial to their ability to spread and cause disease. All *V. fluvialis* strains, with the exception of VP376, possess a full complement of mannose-sensitive hemagglutinin Type IV (MSHA) pilus genes (Online Resource 3). In *V. parahaemolyticus*, this pilus is associated with intestinal colonization by facilitating the secretion of Type III secretion system effector proteins into host cells by causing adhesion to the cell surface (O'Boyle et al. 2013). It also plays a role in mediating cell-rounding cytopathic effects (O'Boyle et al. 2013) and attachment to abiotic surfaces, thereby aiding in the formation of biofilms (Shime-Hattori et al. 2006). All three strains contain genes for a Type IV Tad pilus (Online Resource 3). In *V. vulnificus*, this pilus helps in the formation and persistence of biofilms by promoting initial surface attachment, auto-aggregation, and resistance to mechanical clearance (Pu and Rowe-Magnus 2018). All these could help in the colonization of shellfish.

Moreover, all three *V. fluvialis* strains have genes encoding another type IV pilus present in *V. cholerae*, *V. vulnificus*, and *V. parahaemolyticus*, called ChiRP (chitin-regulated pilus) (Online Resource 3). However, the *pilA* gene which encodes the crucial pilin protein is absent in strains S-5 and S-9 (isolated from river and shrimp farm sediment, respectively). This pilus is associated with the colonization of chitinous surfaces in the environment (Bassler et al. 1991) and may help in the attachment of *V. fluvialis* to shellfish. The presence of three separate type IV pili represents multiple potential mechanisms these strains could use to attach to shellfish and the human intestine. It is worth investigating further to determine whether each pilus is dedicated to the attachment to a particular host.

V. fluvialis also contains *trxA* encoding thioredoxin which is involved in the expression of type IV pili-related genes in *Acinetobacter baumannii* (May et al. 2019). Furthermore, all three strains contain genes encoding the extracellular enzymes chitinase and metalloproteases which are related to the ability of *Vibrio* species to infect and cause disease in shrimp (Hasegawa et al. 2008). The FlgPOT operon encoding flagellar proteins P, O and T was found to be present in all three strains and is important for intestinal colonization by *V. cholerae* (Morris et al. 2008). Moreover, a total of 43 flagella-associated genes were found (Online Resource 3), including those encoding Flagellar synthesis regulators FleN and FleQ which regulate the expression of genes associated with flagellar synthesis (Dasgupta et al. 2000). The flagella of *Vibrio* species play important roles in motility (both in

an aquatic environment and in the host), biofilm formation, colonization, and virulence (Echazarreta and Klose 2019).

Type II and type VI secretion systems: potential role in the delivery of virulence factors and competition against gut microbiota

Pathogenic bacteria utilize secretion systems to secrete effector proteins across cell membranes, thereby delivering virulence factors to the host tissue and competing against the host microbiota. All strains contain multiple genes (Online Resource 3) involved in the Type VI secretion system (T6SS). This secretion system is associated with the secretion of toxins that help bacteria compete against others in the host body and their natural habitat as well as the killing of host immune cells, interfering with immune cell signaling, enhancing host colonization, and aiding in cytotoxicity in other pathogenic *Vibrio* species (Russell et al. 2014a, b; Russell et al. 2014a, b; Hachani et al. 2016).

There are several effector proteins of the T6SS, including the Hcp family of proteins which play a role in bacterial competition and interaction with host cells (Zhou et al. 2012; Peng et al. 2016). Varying copy numbers of *hcp-2* (Hemolysin-Coregulated Protein) were found in our strains. Hcp-2 is involved in competition against other microbes, biofilm formation, and adhesion to host cells (Zhou et al. 2012; Peng et al. 2016; Ding et al. 2018). All strains also have genes for another effector, VgrG-3, which is a toxin that degrades peptidoglycans, thereby aiding in the delivery of the other T6SS toxins of *V. cholerae*, and causing cytolysis (Brooks et al. 2013; Dong et al. 2013; Fu et al. 2013). Multiple genes encoding the effector VgrG were found in most strains, including VgrG-2 which induces autophagy of macrophages by *V. parahaemolyticus* (Yu et al. 2015). Additionally, all three of our strains possess components of a Type II secretion system (T2SS) (Online Resource 3). This system is used by *V. cholerae* to transport virulence factors (Korotkov et al. 2012), including the cholera toxin (Sikora 2013), and the previously discussed PrtV (Rompikuntal et al. 2015).

Biofilm formation

The formation of biofilms is vital to the ability of pathogenic bacteria to survive in the bile-rich environment of the human gastrointestinal tract. Moreover, it is essential to the ability of *Vibrio* species to attach to abiotic surfaces as well as marine organisms. While the phenotypic results of our biofilm formation assay (Online Resource 7) revealed the three strains of this study to form weak biofilms, the propensity of *V. fluvialis* to form biofilms has been demonstrated in another study (Liang et al. 2013). All three strains have genes for a regulatory pair of proteins, the polyamine sensor NspS, involved in biofilm formation, and C-di-GMP

phosphodiesterase MbaA, a repressor of biofilm formation (Karatan et al. 2005). Moreover, these strains contain the *OppABCD* gene cluster (encodes an oligopeptide permease) (Lee et al. 2004) which is involved in biofilm formation (Online Resource 3). Given the presence of genes involved in biofilm synthesis, these strains likely are capable of producing effective biofilms. The method we used to detect biofilm formation was optimized for *E. coli* rather than *Vibrio* spp. and thus may not have favored the formation of biofilm by *V. fluvialis*. This may explain the weak biofilm formation observed in our in vitro experiments.

Analysis of regulatory and other virulence factors

All strains contain numerous factors related to metabolic processes such as energy, lipid, amino acid, iron, and nucleotide metabolism (Online Resources 4 and 9). Several genes involved in the regulation of virulence gene expression were found. They include those encoding the transcriptional activator ToxR (regulates virulence gene expression in *V. cholerae*), Chaperone protein DnaK, Cyclic AMP receptor protein CRP, RNA polymerase-binding transcription factor DksA, Autoinducer-2 production protein LuxS, Stringent starvation protein A sspA, and RNA-binding protein Hfq. Furthermore, *V. fluvialis* has the D-sedoheptulose 7-phosphate isomerase gene *gmhA*, involved in LPS synthesis and with epithelial cell invasion and transepithelial migration of PMN leukocytes by *S. enterica* (May et al. 2019).

All strains contain the gene *zwf* which encodes Glucose-6-phosphate 1-dehydrogenase. This protein is important in overcoming the antimicrobial effects of phagocyte respiratory burst in *S. typhimurium* (Lundberg et al. 1999). Moreover, they also possess the gene encoding translation elongation factor Tu which plays a role in protein synthesis, adherence to host cells, triggering an immune response, and interaction with immune regulators in different pathogens (Harvey et al. 2019).

Antibiotic resistance

Antibiotic resistance is a growing public health concern. Given the unrestricted use of antibiotics in aquaculture, there is potential for the development of antibiotic resistance in microorganisms living on cultured fish and shrimp. Strain S-10 has genes encoding the aminoglycoside inactivating enzymes *APH(6)-Id* and *APH(3'')-Ib* (can be encoded on mobile genetic elements such as plasmids or the chromosome) which is predicted to allow it to be resistant to streptomycin. It also has the *sul2* gene encoding a sulfonamide-resistant dihydropteroate synthase (conferring resistance to sulfamethoxazole) that is commonly found on small plasmids. Moreover, it also has the *tet(35)* gene which confers resistance against tetracycline antibiotics including

doxycycline and tetracycline as predicted by CGI ResFinder. Strain S-10 was also predicted to be resistant to the aminoglycoside antimicrobials florfenicol (used in veterinary medicine, including aquaculture) and chloramphenicol (used through various routes of administration and to treat a variety of infections, including cholera) due to the presence of the *floR* gene (encodes a Phenicol-Specific Efflux Pump) which is usually located in chromosome-borne mobile genetic elements, lending itself to horizontal gene transfer (Online Resource 5). The presence of these genes highlights the potential of shrimp farms being potential sources of *V. fluvialis* carrying mobile antibiotic resistance genes. Strains S-5 and S-9 were predicted to be susceptible to all antimicrobials in the CGI ResFinder database, while CARD annotated 4 acquired resistance genes and 1 resistance-conferring point mutation. The same genes are also present in S-10 (Online Resource 5).

In comparison to our three strains, the clinical isolates from icddr typically contain more antimicrobial resistance genes including the plasmid-borne OXA-10, PER-7, CARB-3, QnrVC4, and ANT(3'')-IIa (Online Resource 10). Among these, the extended-spectrum beta-lactamases OXA-10 and PER-7, and the carbapenemase CARB-3 are of particular public health significance. However, similar to our strains, the environmental strains isolated by icddr contain very few antimicrobial resistance genes, with none of them containing the ones mentioned above. This observation points to a clear distinction between two separate populations of *V. fluvialis* in Bangladesh, clinical versus environmental. Nonetheless, the presence of these acquired antimicrobial resistance genes in the *V. fluvialis* gene pool in Bangladesh represents a potential hazard given the potential of environmental strains, including those found in shrimp farms, to acquire these genes.

Mobile genetic elements

Mobile genetic elements (MGE) such as plasmids and prophages contribute to the genome plasticity of bacteria. Moreover, they are crucial in the ability of bacteria to acquire traits that increase their fitness, helping them adapt to certain niches and survive environmental pressures such as antimicrobials. Classes of MGE that are commonly found in *Vibrio* spp. include plasmids, insertion sequences, and prophage.

Three plasmids were identified in strain S-10 by PLSDb: *Klebsiella pneumoniae* strain KP2442 plasmid pKP2442_7c331, *Klebsiella quasipneumoniae* strain Pinhead Larry plasmid pPinheadLarry_02, and *Acinetobacter baumannii* strain VB11737 plasmid pVB11737_5 (Online Resource 6). The *Acinetobacter baumannii* plasmid contains a gene encoding the previously mentioned sulfonamide-resistant dihydropteroate synthase *Sul2*, while the *Klebsiella*

pneumoniae and *Klebsiella quasipneumoniae* plasmids each contain a gene encoding the previously mentioned chloramphenicol/florfenicol efflux MFS transporter *FloR*. On the other hand, while strains S-5 and S-9 were not found to possess any plasmids by PLSDb, the presence in strains S-5 and S-9 of the IncP-type DNA relaxase *Tral* gene which is typically found on plasmids indicates that they may possess as yet uncharacterized plasmid/s. This highlights the need for the development of plasmid annotation tools that are optimized to detect plasmids in *Vibrio* species.

The insertion sequences of transposable elements were found in all three strains. Strains S-5 and S-9 possess an IS21 family insertion sequence that is similar to ISVch3 of *V. cholerae*. Meanwhile, strain S-10 was found to possess 8 insertion sequences, including the previously mentioned ISVch3. Moreover, the typically plasmid-borne antimicrobial resistance genes *APH(6)-Id* and *APH(3'')-Ib* were found on a Tn3 family transposable element (not a plasmid per PLSDb) that is similar to Tn5393 of *Erwinia amylovora*.

Multiple prophage regions were found in all three strains through the use of PHASTER, with all containing two intact and 1 incomplete regions. Moreover, strain S-10 has 2 questionable hits while S-5 and S-9 have 1 each. The only (intact) region common to all three is most similar to the *Vibrio_K139* phage, whereas S-5 and S-9 also both possess the intact region most similar to the *Vibrio_VP882* phage. Most of the other prophage regions are also similar to *Vibrio* phages while, the incomplete phage regions are more closely related to non-*Vibrio* phages: *Serratia_vB_SmaA_3M* phage in S-5, *Serratia_phiMAM1* phage in S-9, and *Escherichia_500465_1* phage in S-10.

The prophage regions mostly contain genes encoding phage-related proteins and a few genes encoding hypothetical proteins. However, none of these regions contain genes encoding bacterial proteins as predicted by PHASTER.

Nonetheless, the presence of the acquired virulence factors and antimicrobial resistance genes in our three strains (particularly S-10) as discussed prior in this report indicates the ability of *V. fluvialis* organisms isolated from shrimp farm and river sediment to acquire fitness factors through horizontal gene transfer via mobile genetic elements.

Conclusion

To the best of our knowledge, this is the first study that presents a draft-genome analysis of Bangladeshi *V. fluvialis* strains isolated from shrimp farm and river sediments. Our results show that these *V. fluvialis* organisms have many virulence factors in common with other pathogenic *Vibrio* species including *V. cholerae*, mobile genetic elements

including plasmids, transposable elements, prophages, and plasmid-borne and transposon-borne antimicrobial resistance genes. The identification of these genetic elements demonstrates that *V. fluvialis* organisms in shrimp farms and rivers of Bangladesh hold potential as human pathogens and have a propensity to acquire antibiotic resistance genes, making them a potential public health hazard. As such, our results hold significance for the design of strategies to control the spread of enteric disease from shrimp farms and rivers. In addition to *V. cholerae* and other pathogenic species, the monitoring of *V. fluvialis* should also be prioritized.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00203-022-03128-w>.

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Author contributions Conceptualization: Abdul Mueed Ibne Momen, Maqsd Hossain; supervision: Chowdhury Rafiqul Ahsan; methodology: Abdul Mueed Ibne Momen; investigation: Abdul Mueed Ibne Momen, Juthi Biswas; data curation: Abdul Mueed Ibne Momen, Juthi Biswas; formal analysis: Abdul Mueed Ibne Momen, Aura Rahman; visualization: Abdul Mueed Ibne Momen; validation: Maqsd Hossain, Chowdhury Rafiqul Ahsan, Abdul Mueed Ibne Momen; writing—original draft preparation: Abdul Mueed Ibne Momen, Mahmuda Yasmin, Jamalun Nessa, Aura Rahman, Juthi Biswas; writing—review and editing: Abdul Mueed Ibne Momen, Maqsd Hossain, Chowdhury Rafiqul Ahsan.

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Data availability The genome sequences of S-5, S-9, and S-10 have been released in GenBank under the accession numbers QKNU00000000, QJSE00000000, and QJSF00000000, respectively.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical statement This is an observational study. The Ethical Review Committee for Animal Experimentation of the Faculty of Biological Sciences, University of Dhaka has confirmed that no ethical approval is required.

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